

Demonstration of Proposed Features

Date	14 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated(Yes / No)	Remarks
1	Crop Recommendation	Recommends the most suitable crop based on soil nutrients and environmental parameters.	Implemented	Yes	Core functionality completed.
2	Soil Parameter Input	Allows users to enter Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values.	Implemented	Yes	User-friendly input form developed.
3	Machine Learning Prediction	Uses the trained machine learning model (.pkl) to generate crop recommendations.	Implemented	Yes	Successfully integrated with the Flask backend.
4	Input Validation	Validates user inputs before sending them to the prediction model.	Implemented	Yes	Prevents invalid or incomplete inputs.
5	Responsive Web Interface	Provides a clean and responsive interface for easy interaction.	Implemented	Yes	Built using HTML, CSS, and JavaScript.
6	Future Enhancements	Multilingual support	Pending	No	Planned for future versions.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83.3%